STAT 559 – Note 6
Random Variables, Laws, and Independence

Edited transcription of handwritten lecture notes

These notes transcribe and edit the indicated handwritten PDF. Source-page markers preserve traceability. Yellow boxes retain original TODOs; red boxes explain mathematical corrections; blue boxes add explanations or proof details. The reference is *stats310notes.pdf* (provided in `notes/`); references below use its printed section numbers.

Source page(s): 1–2

1 Random variables and generated information

An Ω'-valued random variable on $(\Omega, \mathcal{F}, \mathbb{P})$ is a measurable map into a measurable space $(\Omega', \mathcal{F}')$. We primarily consider real-valued random variables, with target $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$. The event notation is

$$\{X \in A\} = \{\omega : X(\omega) \in A\}, \tag{1}$$

$$\{X \in A, Y \in B\} = \{\omega : X(\omega) \in A,\ Y(\omega) \in B\}. \tag{2}$$

For measurable $g : \mathbb{R} \to \mathbb{R}$, $g(X)$ means the composition $g \circ X$, which is again measurable. The sigma-algebra generated by X is

$$\sigma(X) = \{X^{-1}(A) : A \in \mathcal{B}(\mathbb{R})\}. \tag{3}$$

It is the smallest sigma-algebra on Ω making X measurable. For any index set I, possibly uncountable,

$$\sigma(X_i : i \in I) = \sigma\left(\bigcup_{i \in I} \sigma(X_i)\right). \tag{4}$$

For a fair coin, take $\Omega = \{H, T\}$, $\mathcal{F} = 2^{\Omega}$, $\mathbb{P}(\{H\}) = \mathbb{P}(\{T\}) = 1/2$, and $X(H) = 1, X(T) = 0$. Then $\mathbb{P}(X = 1) = 1/2$.

Source page(s): 3

2 The law of a random variable

The *law* of X is the induced probability measure on the target:

$$\mu_X(A) = \mathbb{P}(X \in A) = \mathbb{P}(X^{-1}(A)), \qquad A \in \mathcal{B}(\mathbb{R}). \tag{5}$$

It is also denoted P_X; the notation $X \sim \mu_X$ specifies this distribution. For example, take the probability space

$$(\Omega, \mathcal{F}, \mathbb{P}) = ([0,1], \mathcal{B}([0,1]), \lambda), \qquad X(\omega) = \omega^2. \tag{6}$$

The sample-space measure is uniform, while the law of X is its pushforward under squaring.

Correction: Where the randomness is represented

The source says that $\mathbb{P}$ has "nothing to do" with the randomness of X. The function X itself is deterministic; its probabilistic behavior is determined jointly by X and $\mathbb{P}$ through μ_X. In this example its cdf on $[0,1]$ is $\sqrt{t}$, not the uniform cdf. No additional randomness axiom is needed (reference Sections 5.1 and 5.3).

Source page(s): 4–5

3 Cumulative distribution functions

Define

$$F_X(t) = \mathbb{P}(X \le t) = \mu_X((-\infty, t]). \tag{7}$$

A function $F : \mathbb{R} \to [0,1]$ is a cdf exactly when it is nondecreasing, right-continuous, and satisfies

$$\lim_{t \to -\infty} F(t) = 0, \qquad \lim_{t \to \infty} F(t) = 1. \tag{8}$$

Correction: The missing monotonicity condition

The source leaves condition (a) blank. Nondecreasingness is required, in addition to right-continuity and the two limits. The source calls this a Skorokhod theorem; here it is the inverse-cdf construction of reference Proposition 5.2.2, distinct from the later Skorokhod representation theorem for weak convergence.

Given such F, take $\Omega = (0,1)$ with Lebesgue probability measure and define

$$X(u) = F^{-1}(u) := \inf\{t \in \mathbb{R} : F(t) \ge u\}. \tag{9}$$

The endpoint limits make this finite for $0 < u < 1$. Monotonicity and right-continuity yield $X(u) \le t$ if and only if $u \le F(t)$. Thus X is measurable and $\mathbb{P}(X \le t) = F(t)$. Conversely monotonicity of probabilities and continuity of a measure along increasing and decreasing sequences establish the cdf properties.

Correction: Avoiding inverse-cdf endpoint ambiguities

The source uses $[0,1]$. The infimum formula can produce infinite values at zero or one. Using $(0,1)$, as in the reference, avoids this; alternatively assign arbitrary finite values at these two null endpoints.

Source page(s): 5–6

4 The cdf determines the law

Two real random variables have the same cdf if and only if they have the same law. One direction follows by evaluating the laws on $(-\infty, t]$. For the other define

$$\mathcal{D} = \{(-\infty, t] : t \in \mathbb{R}\}, \qquad \mathcal{G} = \{A \in \mathcal{B}(\mathbb{R}) : \mu_X(A) = \mu_Y(A)\}. \tag{10}$$

Then $\mathcal{D}$ is a pi-system generating $\mathcal{B}(\mathbb{R})$ and equal cdfs imply $\mathcal{D} \subseteq \mathcal{G}$.

Original TODO: Check the lambda-system step

The whole space lies in $\mathcal{G}$ since both laws have mass one. If $A \in \mathcal{G}$, then $\mu_X(A^c) = 1-\mu_X(A) = 1 - \mu_Y(A) = \mu_Y(A^c)$. Countable additivity gives closure under disjoint countable unions. Hence $\mathcal{G}$ is a lambda-system, and Dynkin's theorem proves equality on all Borel sets (reference Proposition 5.3.1).

Source page(s): 7–8

5 Densities and absolute continuity

For two measures, $\nu \ll \mu$ means $\mu(A) = 0 \Longrightarrow \nu(A) = 0$ for every measurable A. A probability law admitting a density with respect to Lebesgue measure satisfies

$$\mu_X(A) = \int_A f \, d\lambda, \qquad f \geq 0, \quad \int_{\mathbb{R}} f \, d\lambda = 1. \tag{11}$$

The density is denoted $d\mu_X/d\lambda$, a Radon–Nikodym derivative. Conversely such a nonnegative measurable normalized f defines a probability law by this formula. Reference Section 5.4 develops the density definition directly.

Supplement: The source's optional decomposition remark

For a finite measure ν on $\mathbb{R}$, the Lebesgue decomposition relative to λ has the form $\nu = \nu_{\text{ac}} + \nu_{\text{s}}$ with $\nu_{\text{ac}} \ll \lambda$ and $\nu_{\text{s}} \perp \lambda$. Singularity means the singular part is concentrated on a Lebesgue-null measurable set. This is a side remark in the handwriting, not proved in the provided reference's density section. The unrestricted phrase "every measure" in the source requires general measure-theoretic hypotheses; the finite-law case is the one relevant here.

To verify a candidate density it suffices to check its integral formula on closed intervals $[a, b]$ whose endpoints are continuity points of F_X. With f a nonnegative measurable candidate, increasing such intervals to $\mathbb{R}$ first shows normalization. Continuity-point approximation extends the equality to arbitrary intervals, then the pi–lambda theorem gives all Borel sets (reference Proposition 5.4.1).

Source page(s): 9

Supplement: Density homework

If f, g are densities for the same law, $\int_A (f - g)\, d\lambda = 0$ for every Borel A. Applying this to $\{f > g\}$ and $\{f < g\}$ proves $f = g$ a.e. Also the cdf associated to f is

$$F(t) = \int_{-\infty}^{t} f(y)\, d\lambda(y). \tag{12}$$

The integral is unchanged by including the endpoint because singletons are Lebesgue-null. See reference Exercises 5.4.2–5.4.3. The source assigns Sections 5.5–5.6 (standard continuous and discrete distributions) as self-reading.

Source page(s): 10–11

6 Expectation and changing to the law

Expectation is a Lebesgue integral:

$$\mathbb{E}X = \int_\Omega X(\omega)\, d\mathbb{P}(\omega), \tag{13}$$

provided the positive and negative parts do not both have infinite integral. The Cauchy distribution is an example with undefined mean. For $X \sim N(m, \sigma^2)$ the expectation is m. The *law of the unconscious statistician* states that for Borel g, whenever the integral is defined,

$$\mathbb{E}[g(X)] = \int_\Omega g(X(\omega))\, d\mathbb{P}(\omega) = \int_{\mathbb{R}} g(x)\, d\mu_X(x). \tag{14}$$

If μ_X has density f_X, this equals $\int_{\mathbb{R}} g(x) f_X(x)\, d\lambda(x)$.

Supplement: Why this change of space is valid

For indicator $g = \mathbf{1}_A$ this is the definition of μ_X. Extend by finite sums to nonnegative simple functions, by MCT to nonnegative measurable functions, then by positive and negative parts to signed functions with defined integral (reference Section 6.1).

Supplement: Tail-integral homework

For nonnegative X, Tonelli applied to $\mathbf{1}_{\{0<t<X(\omega)\}}$ gives

$$\mathbb{E}X = \int_0^\infty \mathbb{P}(X > t)\, dt. \tag{15}$$

Indeed $X(\omega) = \int_0^\infty \mathbf{1}_{\{t<X(\omega)\}}\, dt$. The source writes $\mathbb{P}(X \geq t)$; that has the same integral because the distribution has only countably many atoms. Both formulas allow infinity (reference Section 6.1).

The source emphasizes studying the *characteristic function* for the final. Its definition is $\varphi_X(t) = \mathbb{E}(e^{itX})$, and it always exists because the integrand has absolute value one (reference Section 6.4).

Source page(s): 12–13

7 Independence of sigma-algebras, events, and variables

Sub-sigma-algebras $\mathcal{F}_1, \ldots, \mathcal{F}_n \subseteq \mathcal{F}$ are independent if for every $A_i \in \mathcal{F}_i$,

$$\mathbb{P}\left(\bigcap_{i=1}^n A_i\right) = \prod_{i=1}^n \mathbb{P}(A_i). \tag{16}$$

Events $A_1, \ldots, A_n$ are independent when their generated sigma-algebras are independent. Explicitly,

$$\sigma(A_i) = \{\varnothing, \Omega, A_i, A_i^c\}, \tag{17}$$

with duplicate sets removed. An arbitrary family of sigma-algebras is independent if every finite subfamily is independent. Random variables $\{X_i\}_{i\in I}$ are independent if $\{\sigma(X_i)\}_{i\in I}$ are independent.

Original TODO: Why is sigma(X) interpreted as information?

Its events are precisely the events of the form $\{X \in B\}$ that can be determined by observing X. For the coin example these distinguish heads from tails; a constant random variable generates only $\{\varnothing, \Omega\}$. Independence of these sigma-algebras formalizes that any finitely many observable events across the separate variables have probabilities that factor. See reference Sections 5.1 and 7.1.

Source page(s): 14–15

8 Connecting the elementary definition and Borel–Cantelli

For Bernoulli variables, independence can be checked by factoring all four joint point masses:

$$\mathbb{P}(X = x, Y = y) = \mathbb{P}(X = x)\mathbb{P}(Y = y), \qquad x, y \in \{0, 1\}. \tag{18}$$

Summing over the relevant values gives the event-based definition. In general the equivalent statement is that the joint law is the product of the marginal laws; the symbols $P(X, Y) = P(X)P(Y)$ in the source abbreviate this measure identity. The first Borel–Cantelli lemma says

$$\sum_n \mathbb{P}(A_n) < \infty \quad \Longrightarrow \quad \mathbb{P}(A_n \text{ infinitely often}) = 0. \tag{19}$$

The second says that for independent events,

$$\sum_n \mathbb{P}(A_n) = \infty \quad \Longrightarrow \quad \mathbb{P}(A_n \text{ infinitely often}) = 1. \tag{20}$$

Supplement: Proof of the second lemma

For fixed $m \le n$, independence and $1 - u \le e^{-u}$ imply

$$\mathbb{P}\left(\bigcap_{k=m}^{n} A_k^c\right) = \prod_{k=m}^{n} (1 - \mathbb{P}(A_k)) \le \exp\left(-\sum_{k=m}^{n} \mathbb{P}(A_k)\right) \longrightarrow 0. \tag{21}$$

Hence the probability of no occurrences after any fixed m is zero. The event of finitely many occurrences is the countable union of these null events, so it is null (reference Theorem 7.3.1).

Correction: Independence is needed for the zero–one conclusion here

Together the two lemmas show that for an independent sequence, the infinitely-often event has probability zero or one according to convergence or divergence of the probability sum. Without independence it may have any probability: taking $A_n = A$ for every n gives probability $\mathbb{P}(A)$. Page 14's sum must refer to the same probability measure as its conclusion, rather than an unrelated μ.

Source page(s): 16

9 Optional topics recorded in the source

The final page lists the Kolmogorov zero–one law and the construction of an infinite product $P_1 \otimes P_2 \otimes \cdots$ as optional topics. The supplied reference treats these in Sections 7.4 and 3.3 respectively. The source contains no proof on this page.